

Bahria University

Karachi Campus



LAB EXPERIMENT NO.

13

LIST OF TASKS

TASK NO	OBJECTIVE
1	Develop an AI agent that analyzes and summarizes documents, ensuring data privacy using openai-agent sdk or any opensource (if openai-api-key is not available).

Submitted On:

Date: _____

Task No.01: Develop an AI agent that analyzes and summarizes documents, ensuring data privacy using openai-agent sdk or any opensource (if openai-api-key is not available).

Solution:

```
from llama_cpp import Llama

llm = Llama(model_path="model.gguf", n_ctx=2048, n_threads=4)
```

llama_model_loader: loaded meta data with 20 key-value pairs and 291 tensors from model.gguf (version GGUF V2)
llama_model_loader: Dumping metadata keys/values. Note: KV overrides do not apply in this output.

llama_model_loader: - kv 0:	general.architecture str	= llama
llama_model_loader: - kv 1:	general.name str	= mistralai_mistral-7b-instruct-v0.1
llama_model_loader: - kv 2:	llama.context_length u32	= 32768
llama_model_loader: - kv 3:	llama.embedding_length u32	= 4096
llama_model_loader: - kv 4:	llama.block_count u32	= 32
llama_model_loader: - kv 5:	llama.feed_forward_length u32	= 14336
llama_model_loader: - kv 6:	llama rope-dimensions u32	= 128

```
def summarize_text(text):
    prompt = f"Summarize this text:\n{text}\n\nSummary:"
    output = llm(prompt, max_tokens=300, stop=["\n"])
    return output["choices"][0]["text"].strip()
```

```
[ ] import pdfplumber
from google.colab import files

uploaded = files.upload()
pdf_path = list(uploaded.keys())[0]

def extract_text_from_pdf(path):
    all_text = ""
    with pdfplumber.open(path) as pdf:
        for page in pdf.pages:
            page_text = page.extract_text()
            if page_text:
                all_text += page_text + "\n"
    return all_text

doc_text = extract_text_from_pdf(pdf_path)
print("✅ Text extracted from PDF.")
```

Choose Files sample.pdf

- sample.pdf(application/pdf) - 1453 bytes, last modified: 5/22/2025 - 100% done

WARNING:pdfminer.pdfpage:CropBox missing from /Page, defaulting to MediaBox
WARNING:pdfminer.pdfpage:CropBox missing from /Page, defaulting to MediaBox
Saving sample.pdf to sample.pdf
✅ Text extracted from PDF.

```
chunk = doc_text[:2000]
summary = summarize_text(chunk)

print("🔥 Summary:\n")
print(summary)
```

llama_perf_context_print: load time = 54100.03 ms
llama_perf_context_print: prompt eval time = 54099.30 ms / 161 tokens (336.02 ms per token, 2.98 tokens per second)
llama_perf_context_print: eval time = 30923.30 ms / 51 runs (606.34 ms per token, 1.65 tokens per second)
llama_perf_context_print: total time = 85054.22 ms / 212 tokens

🔥 Summary:

This PDF document is a test for Retrieval-Augmented Generation (RAG) systems, containing multiple sentences on topics such as Artificial Intelligence, Machine Learning, Python, and libraries used in Natural